The Rule-First Study: do ancient Vedic earthquake rules match real earthquakes?

In plain words. Old Indian texts say that the positions of the Moon, Sun and planets can hint at when and where earthquakes happen. This report takes those rules one by one and checks them against real, deadly earthquakes, and then against thousands more. It is called the Rule-First Study because it starts from the rules and looks for matching earthquakes; the companion Data-First Study does the reverse. Technical terms are kept where they help, and each is explained in the Glossary (section 18). Report date: 20 October 2026.

Sources. Parāśara Tantra (PT; R N Iyengar edition), Brihat Saṃhitā (BS; Sanskrit scan, cross-checked against a supplied English translation), Cunningham's Ancient Geography of India, Madhavan's Muddle of Ayanamsa, and the TG-SAI GitHub repository (github.com/karthikeyadarbha/AI-Catastrophe-Analytics).

The answer in brief

No rule, sky measure or model tested here predicted earthquakes better than chance. The table gives each question, the answer and where to find the evidence.

Question	Answer	Where
Do the Deadly 15 earthquakes follow the Vedic patterns?	No. 71 tests, none with p below 0.05 (smallest 0.059).	Section 4
Do the patterns hold on thousands of earthquakes?	No. 23 rules frozen in advance; none passed on the holdout (best p=0.039 against a bar of 0.0022).	Section 5
Do the two ancient texts agree, and does either hold up?	They agree on 23 of 27 nakṣatras; both are null. The Sanskrit and English readings of the Brihat Saṃhitā agree except for one printing slip that changes nothing.	Section 6
Does the crescent-Moon 'carriage pole' omen work when worked out exactly?	No, in every era.	Section 7
Do the rules work inside particular earthquake belts?	No. A faint, uncorrected crescent pattern in Japan and Indonesia needs new earthquakes to test.	Section 8
Does the TG-SAI repository work?	No. AUC about 0.50 once its flaws are fixed, and 0.50 to 0.51 on three older data sets.	Section 9
Does verse 18.7 (repeating tremors) or the comet verse hold?	No excess over ordinary aftershock decay; no comet excess.	Section 10
Does the choice of ayanāṃśa change the answer?	No. Named systems and a 29-offset sweep stay null.	Section 11
Do older earthquakes (1900–1989) rescue any lead?	No. The one lead closed.	Section 12

1. Study design and how to read the numbers

Purpose. To find out whether the testable claims in the old texts, and the physics measures in the TG-SAI code, are linked to real earthquakes. **Approach.** For each earthquake we work out where the Sun, Moon and planets stood. For each rule we count how many earthquakes fit it and how many would fit by chance, using a background of ordinary moments in the same years. A rule counts only if earthquakes fit it clearly more often than chance.

Data sets (names used in both reports)

Name	Years	How selected	Events	Used for
Deadly 15	1999–2026	The 15 deadliest earthquakes in the USGS catalogue (file usgs_catalogue_15_events.csv)	15	First tests of every rule
Modern M6.5+ set	1990–2026	All M6.5+ earthquakes, aftershocks removed (1,709 raw)	1,270 (632 before 2008 = discovery; 638 from 2008 = holdout)	Pre-registered 23 rules; first data-first scan
Modern M5.5+ set	1990–2026	All M5.5+ earthquakes, aftershocks removed; includes the Modern M6.5+ set	9,368 (4,661 before 2008; 4,707 from 2008); 8,098 are not in the Modern M6.5+ set	Larger-sample replication; gap tests
Older set 1	1964–1989	M5.5+, aftershocks removed (11,329 raw)	6,111	Replication on never-used data
Older set 2	1900–1963	M7.0+, aftershocks removed (759 raw)	670	Replication on never-used data

Name	Years	How selected	Events	Used for
Repository file	1850–1950	Table of historical M5+ quakes shipped with the TG-SAI repository	5,003	Scoring the TG-SAI model
Aftershock set	1990–2026	481 main shocks of M7.0+ with USGS aftershocks (M5.0+, 100–300 km, 0–90 days)	481 sequences (11,608 records)	Verse 18.7 robustness test

Positions come from the astral-ephemeris tool for the Deadly 15 and the Modern M6.5+ set, and from our own formulas for the other sets. Our formulas were checked against the tool on the Modern M6.5+ set: Moon and Sun agree to 0.01°, planets within 0.17° at worst, and every retrograde flag is identical. The sidereal frame uses the Lahiri ayanāṃśa (23.85° at 2000, growing 0.014° a year, so about 23.7° in 1990 and 24.2° in 2026); the tropical frame is sidereal plus that value. Eclipses use our own Meeus-based finder (about ±1.4 days). Statistics are binomial and Monte Carlo baselines, upper-tail enrichment p-values, and Spearman and Kruskal–Wallis tests for severity. Chapter text was read from OCR of the scanned PDFs and key verses were checked on page images.

What a p-value is, and what it must be to count

Meaning. A p-value answers one question: if nothing but chance were at work, how often would a result at least this extreme appear? Take the sharp-nakṣatra rule: on the holdout, 111 of 638 events had the Moon in the sharp group, against 94.5 expected from the background grid. p=0.039 means that if the Moon's position were unrelated to earthquakes, about 4 random sets of 638 events in 100 would show 111 or more. It is not the probability that the pattern is real, and it does not measure the size of the effect (here a 17% excess).

Why 0.05 is not enough here. A p below 0.05 is the usual bar for one pre-planned test, because chance alone passes it 1 time in 20. This study ran 23 rules, so about one rule would be expected to dip below 0.05 even if every rule were false (23 × 0.05 ≈ 1.2). To keep the chance of any false positive across the whole family at 5%, each rule must meet the Bonferroni threshold: 0.05 ÷ 23 = 0.0022, about 1 in 460. The sharp-group result at p=0.039 is 18 times above that bar. The smallest holdout p-value among all 23 rules is therefore what a family of true nulls would typically produce.

The second condition: replication. The rules were fixed before the data were split. A rule counts as supported only if it (1) passes 0.0022 on the holdout (2008 onward) and (2) shows the same direction in the discovery half (before 2008). The sharp group fails both: it misses the threshold, and in the discovery half it shows 91 events against 93.6 expected (a slight deficit, p=0.63). A real effect should appear in both halves; a chance cluster usually does not.

How to read the tables in this report. Large p (near 0.5 or above) means the data look like chance; values such as 0.9 or 0.97 are not evidence against a pattern in the other direction, only that observed and expected counts were close. p between 0.05 and 0.0022 is a hint that needs new data, not a finding. A p-value also says nothing about whether the rule is correct as scholarship: a failure to reject chance means these tests, at this sample size, saw no effect; at the corrected threshold, 638 holdout events would reliably (80% of the time) detect an excess of roughly 35% in a rule like the sharp group, and would miss smaller effects, such as the 17% seen.

2. The source repository and the Deadly 15

The GitHub repository. The study began with a public code project, AI-Catastrophe-Analytics (github.com/karthikeyadarbha/AI-Catastrophe-Analytics), which contains a model called **TG-SAI** (“Topocentric Gravitational-Seismic AI”). In simple terms, it is a computer program that tries to guess how risky a given place and day are for a big earthquake, using only the positions of the Moon, Sun and planets. It looks at seven sky measures: how closely the Sun, Moon and Earth line up (a new or full moon), how Saturn, Jupiter and Mars line up, how close the Moon is to Earth, and how strongly the Moon and Sun sit overhead at the place in question. A machine-learning method (a RandomForest, which is a crowd of simple decision trees voting together) learns from past earthquakes and from 10,000 deliberately chosen ‘non-earthquake’ moments (called hard negatives). The repository then scans a map of fault zones in half-degree squares for 2026–2031 and labels each spot with a risk level: below 50% normal (for example Bhuj 2001), 75–85% high (Sumatra 2004), 85% and above extreme (Tōhoku 2011). The zip we received contained the README, two helper scripts for planet positions and a table of historical M5+ quakes (1850–1950). The main training file it mentions was not in the first zip, so we first rebuilt the seven measures from the repository's own formulas; the file was supplied later and is audited and tested in section 9.4.

The Deadly 15. To judge the repository and the ancient rules on the same footing, we picked the 15 deadliest earthquakes from 1999 to 2026 in the US Geological Survey (USGS) catalogue (file

usgs_catalogue_15_events.csv). For each one we computed the seven repository measures, the Moon's star position (nakṣatra) and the Parāśara Tantra earthquake type (mandala). The table shows what the repository's own formulas predicted for each event. A forecast probability of 50% or more would count as a ‘hit’.

Earthquake	Magnitude	Deaths	TG-SAI probability (repository formulas)	Hit at 50%?
Sumatra 2004	9.1	230,000	79.6%	Yes
Haiti 2010	7.0	220,000	25.8%	No
Sichuan 2008	7.9	87,587	50.7%	Yes
Kashmir 2005	7.6	73,000	70.1%	Yes
Turkey-Syria 2023	7.8	59,500	71.4%	Yes
Bam 2003	6.6	20,000	42.4%	No
Bhuj 2001	7.7	20,000	46.3%	No
Tohoku 2011	9.0	19,700	85.2%	Yes
Izmit 1999	7.6	17,480	61.0%	Yes
Nepal 2015	7.8	8,964	67.6%	Yes
Venezuela 2026	7.5	6,509	15.3%	No
Java 2006	6.3	5,700	83.7%	Yes
Sulawesi 2018	7.5	4,300	45.9%	No
Myanmar 2025	7.7	3,700	26.7%	No
Morocco 2023	6.9	2,960	30.4%	No

A short talk about what this table says. The model looks impressive on a few famous events: Tōhoku (85%), Java 2006 (84%) and Sumatra (80%) score high. But it also gives only 15% to Venezuela 2026, 26% to Haiti 2010 (220,000 deaths) and 27% to Myanmar 2025, and 8 of the 15 sit at or above 50%, which is roughly what a coin toss would give. When we later checked the code, we found why the numbers look better than they are: several of the seven measures sit near their maximum for about a week around every new or full moon, so they say ‘high’ on a large share of all days, and the ‘non-earthquake’ examples were sampled differently from the earthquakes. When we fixed these problems and retrained (section 9), the model's accuracy score (AUC) fell to about 0.50, which is exactly what guessing would give. So the repository is the starting hypothesis, not a proven tool.

3. The rules and where they come from

Each row names the passage, what it says, and how it was turned into a measurable rule. Verse numbers are given where they were read and checked; where only the chapter was used, the chapter is cited. Results are in the sections listed at the end of this section.

Source and passage	What the text says	What was adapted for the experiment
PT 18.2	Indra clipped the wings of flying hills; quakes now arise from four causes	Framing only: four types Anila, Anala, Indra, Varuṇa
PT 18.3–18.6 (Sanskrit checked on page images)	Moon's nakṣatra sets the type; lists of nakṣatras, effects, diseases, regions per type	Moon sidereal nakṣatra → type; corrected lists (6/8/7/6); effects not testable
PT 18 (four parts of day and night)	Quakes in the four parts of day and night correspond to the four types	Local solar hour → type; matched to nakṣatra type
PT 18.7	Repeated tremors at 3, 4, 7 nights, a fortnight, a month, a month and a half portend a king's fall	Lags 3, 4, 7, 14, 30, 45 days from the main shock, and gaps between aftershocks; ±0.25/0.5/1 day; Omori decay baseline; both readings
PT 5 (eclipses), esp. 5.4, 5.7	5.4: eclipse nakṣatra → affected region; 5.7: an earthquake foretells an eclipse in the 24th month	Eclipse within ±7/15/30 days; eclipse 24±1 months after; region match with Cunningham table
PT 11.19 (comets)	An earthquake is a precursor of a comet	Great-comet perihelion within 12 or 24 months of a quake (seven comets)
PT Ch.14 (nakṣatra groups, afflictions)	Dhruva, Mṛdu, Ugra, Tīkṣṇa, Kṣipra, Cara, Miśra; afflicted by planets	Moon in the sharp group (Ārdrā, Āśleṣā, Jyeṣṭhā, Mūla); Moon within 5° of a planet
PT Ch.15 (planetary wars, formations)	Planets close together or in one nakṣatra form named configurations	Pairs within 1° and 3°; three or more planets in one nakṣatra
PT Ch.16 (tithi, karaṇa, muhūrta)	Tithi types, Viṣṭi karaṇa harsh, twilights to avoid	Elongation → tithi and karaṇa; ±48 minutes of sunrise or sunset
PT Ch.3 (Sun)	Six ṛtus of 60° from the winter solstice (Dhaniṣṭhā)	Sun's season in sidereal and tropical frames

Source and passage	What the text says	What was adapted for the experiment
PT Ch.4 (Moon)	A 'yoke' crescent portends earthquakes	Proxy: Moon's ecliptic latitude, north tip raised, waxing-crescent phase
PT Ch.6–10 (planets)	Retrogrades, station nakṣatras, Venus's six circles, Jupiter–Saturn in Kṛttikā/Viśākhā	Retro flags, Venus circle, same-nakṣatra flags
PT Ch.19 (portents)	Haloes, sky brightening before quakes	None possible: no observation records
BS 32.1–2	Rival causes of quakes	Framing only
BS 32.3–7	Day and night in four parts; Gārgya's halves of day and night	Local solar time in 06–12, 12–18, 18–24, 00–06 → Vāyu, Agni, Indra, Varuṇa
BS 32.8–11 (Vāyu), 12–15 (Agni), 16–19 (Indra), 20–22 (Varuṇa)	Seven nakṣatras per type with omens seven days before and effects	Moon's nakṣatra → type; four-class chi-square; Vāyu rule alone
BS 32.23	Result within six months; nirghāta within two	Not tested: needs outcome data
Cunningham, Ancient Geography of India	Ancient regions and their nakṣatra and eclipse links	Region table with grades A/B/C; modern places
TG-SAI repository	Seven physics features, RandomForest	Recomputed with matched negatives, instantaneous and true geocentric features
Madhavan, Muddle of Ayanamsa	Competing ayanāṃśa systems, coincidence years and rates	Named systems and a 29-offset sweep of the Moon's longitude

Which parts are adaptation, not the text. The thresholds (±7/15/30 days, ±0.5 day, 5° and 3° separations), the Gārgya half-day clock time, the Omori baseline, the crescent proxy and the choice of Lahiri were our choices; the verses give the nakṣatra lists, the lags and the qualitative rules. Where the printed English and the Sanskrit differed (PT 18.3–18.5), the Sanskrit was followed.

Where results are. Deadly 15: section 4. Modern M6.5+ set: section 5. Mandala systems and the English cross-check: section 6. Crescent: section 7. Regions: section 8. TG-SAI: section 9. Verse 18.7 and comets: section 10. Ayanāṃśa: section 11. Older eras and the larger samples: section 12 and the Data-First Study.

4. Results on the Deadly 15

The Deadly 15 are: Sumatra 2004, Haiti 2010, Sichuan 2008, Kashmir 2005, Turkey–Syria 2023, Bam 2003, Bhuj 2001, Tōhoku 2011, İzmit 1999, Nepal 2015, Venezuela 2026, Java 2006, Sulawesi 2018, Myanmar 2025, Morocco 2023 (M6.3–9.1; 2,960–230,000 deaths). Positions: astral-ephemeris tool (Lahiri sidereal; sidereal frame = Lahiri ayanāṃśa, 23.85° at 2000 growing 0.014° a year, so about 23.7° in 1990 and 24.2° in 2026; tropical = sidereal + that value). Eclipses: own Meeus-based finder (≈±1.4 d timing). Baselines: binomial and Monte Carlo (random dates/positions). Statistics: upper-tail enrichment p, Spearman and Kruskal–Wallis for severity (deaths, magnitude). Chapter text was read from OCR of the scanned PDF and key verses checked on page images. Later sections repeat the tests on larger sets with rules frozen before testing.

Event	UTC	Mag	Deaths	Moon nakṣatra	PT mandala (Ch.18)
Sumatra 2004	2004-12-26 00:58	9.1	230,000	Mrigashira	Indra
Haiti 2010	2010-01-12 21:53	7.0	220,000	Mula	Varuna
Sichuan 2008	2008-05-12 06:28	7.9	87,587	Ashlesha	Varuna
Kashmir 2005	2005-10-08 03:50	7.6	73,000	Jyeshtha	Indra
Turkey-Syria 2023	2023-02-06 01:17	7.8	59,500	Ashlesha	Varuna
Bam 2003	2003-12-26 01:56	6.6	20,000	Shravana	Indra
Bhuj 2001	2001-01-26 03:16	7.7	20,000	Dhanishta	Anila
Tohoku 2011	2011-03-11 05:46	9.0	19,700	Krittika	Anala
Izmit 1999	1999-08-17 00:01	7.6	17,480	Svati	Anila
Nepal 2015	2015-04-25 06:11	7.8	8,964	Punarvasu	Indra
Venezuela 2026	2026-06-24 22:05	7.5	6,509	Svati	Anila
Java 2006	2006-05-26 22:53	6.3	5,700	Krittika	Anala
Sulawesi 2018	2018-09-28 10:02	7.5	4,300	Bharani	Anala
Myanmar 2025	2025-03-28 06:20	7.7	3,700	Purva Bhadrapada	Anala
Morocco 2023	2023-09-08 22:11	6.9	2,960	Ardra	Varuna

In plain words. for each earthquake we worked out where the Sun, Moon and planets stood, then asked, for every rule, how many of the 15 quakes fit it, and how many would fit by pure chance. A rule only counts if the quakes fit it clearly more often than chance.

Source	Rule (paraphrased)	Operationalisation	Result
Ch.18 Earthquake	Mandala (Anila, Anala, Indra, Varuṇa) set by Moon's nakṣatra, with a day/night split	Moon nakṣatra → mandala; also user's Anila nakṣatra list (Svāti, Abhijit, Dhaniṣṭhā, Aśvinī, P.Phalgunī, Hasta, Citrā)	Anila list: 3 events (Bhuj, İzmit, Venezuela) vs ≈3.3 expected. PT mandalas 4/4/3/4 across the four. Nakṣatra-vs-time-of-day agreement 3/15 (≈3.75 expected). No link to deaths or magnitude.
Brihat Saṃhitā ch.32	Four-circle (maṇḍala) model by nakṣatra	BS mandala per event	Vāyu 4, Varuṇa 4, Indra 3, Agni 4: uniform. Vāyu rule (Moon in 7 nakṣatras, optionally with 03–09h local) on the Modern M6.5+ set (1,270 events): 171 vs 165 and 43 vs 41 expected on the holdout (p=0.32, 0.42); full tests in section 6.
TG-SAI repo	7 features: syzygy, Saturn, Jupiter, Mars alignment, perigee, lunar and solar zenith; RandomForest	Features computed per event (repo's own formulas)	No significant relationship. Features are near-saturated by construction (see section 9).
Ch.5 Rāhu (eclipses)	Eclipse portents; 5.4 nakṣatra→region; 5.7 an earthquake foretells an eclipse in the 24th month	Eclipse within ±7/15/30 d; eclipse 24±1 months after; region match	4 / 4 / 6 of 15 vs 2.6 / 4.0 / 6.6 expected (p 0.25, 0.60, 0.72). 24-month rule: 2 vs 6.6 expected (fewer than chance). Region match: 0 of 4 in-scope events (0.2 expected).
Cunningham geography	Ancient regions (Kāśmīra, Surāṣṭra, Sindhu, Kamboja, Yavana, …) mapped to modern places	Region table (CSV) with source grades A/B/C	Only 4 of 15 events fall in the ancient Indian world; none matched an eclipse region.
Ch.16 Tithi/Karaṇa/Muhūrta	Tithis Nandā…Pūrṇā; 11 karaṇas (Viṣṭi harsh); avoid twilights	Elongation→tithi, karaṇa; ±48 min of sunrise/sunset	All near chance: Riktā 3 vs 3; Viṣṭi 1 vs 2; twilight 4 vs 2 (p=0.13).
Ch.14 Nakṣatra afflictions	Nakṣatra groups (fixed, sharp, fierce, swift, mixed, moving, soft); afflicted by Moon/planet approach	Moon-nakṣatra group; Moon within 5° of a planet	Sharp group (Ardra, Āśleṣā, Jyeṣṭhā, Mūla): 5 vs 2.2 (p=0.059, not significant after correction). Moon–planet ≤5°: 4 vs 2.3 (p=0.18).
Ch.15 Planet conjunctions	Planetary wars; formations (Śṛṅgāṭaka, Vyathā…) from planets in one nakṣatra	Pairs ≤1°/≤3°; 3+ of 5 planets in one nakṣatra	≤1°: 0 vs 0.8; ≤3°: 4 vs 2.3 (p=0.20); cluster 0 vs 0.2.
Ch.3 Sun	Six ṛtus of 4½ equal divisions from winter solstice (Dhaniṣṭhā)	Sun's season, sidereal and tropical frames	Same five Dec–Feb events cluster in both frames: 5 vs 2.5 (p=0.09).
Ch.4 Moon	“Yoke” crescent portends earthquakes; Moon-split-by-planet regions	Proxy: Moon ecliptic latitude (north tip raised) in the waxing-crescent phase, Modern M6.5+ set (section 7)	Holdout 42 of 67 vs 33.6 expected, p=0.026 (needs ≤0.0125); discovery p=0.29. Not supported.
Ch.6–10 Mars→Saturn	Retrogrades, station nakṣatras, Venus's six circles, Saturn–Jupiter in Kṛttikā/Viśākhā	Retro flags; Venus circle; Jupiter–Saturn same nakṣatra	Retro: Mars 2, Mercury 4, Jupiter 3, Venus 2, Saturn 5 (all near expectation). Venus circles within chance in both frames. Jupiter–Saturn same star: Bhuj 2001 only (1 vs 0.6).
Ch.11 Comets, Ch.19 Meteors/haloes	Earthquake precedes a comet (11.19); sky brightening before quakes	Comets: seven great comets 1990–2026. Haloes and sky brightening: no structured records exist	Comets: no excess (p=0.81 at 12 months, 0.41 at 24). Sky portents: untestable (section 13).

Overall. Across 71 enrichment tests, none reached p<0.05 (smallest 0.059). Of about 40 severity correlations, the best (p≈0.02) is what chance produces among that many tests and does not survive correction for multiple testing.

Sidereal and tropical frames

Nakṣatra-based features were computed in both frames. Frame-dependent excesses did not carry over: the Moon-in-sharp-group excess (5 events) is present only sidereally; Venus in the Tiksna circle rises from 3 to 5 tropically (p=0.13); the Dec–Feb season cluster appears in both frames because it is a calendar effect of the same five events. Frame-independent features (tithi, karaṇa, Moon–Sun elongation, planet separations, retrogrades) are unaffected.

Severity correlations

Spearman correlations of planet retrograde count, combust count, same-nakṣatra planet clusters, and Moon/Mars nakṣatra with log(deaths) and magnitude were all weak. The only nominal hits were the largest same-nakṣatra planet cluster vs. magnitude (ρ=0.60 sidereal, p=0.018; 0.57 tropical, p=0.028), with only three distinct cluster values and about 20 correlations tried; after correction p≈0.36. Combust-planet count vs. deaths: ρ=−0.39 (p=0.15). Eclipse distance vs. deaths: 0.10.

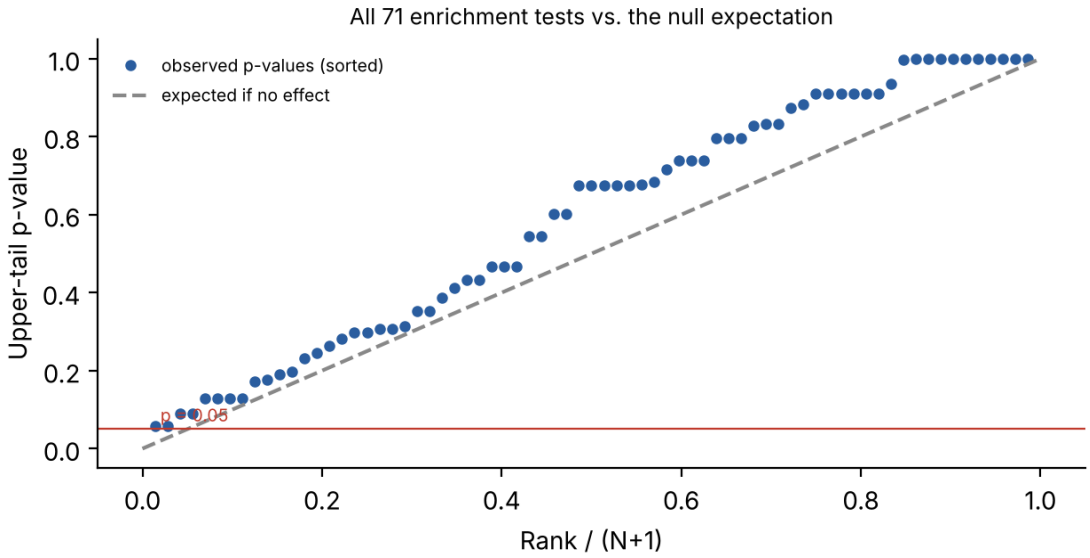


Figure 1. Sorted p-values from all 71 enrichment tests. Points on the dashed line are what pure chance looks like; none falls below 0.05.

Why the sample was extended. The Deadly 15 can detect only a very large effect, so every rule was repeated on the Modern M6.5+ set (section 5) and then on larger and older sets (section 12 and the Data-First Study).

5. The 23 pre-registered rules on the Modern M6.5+ set

Data. The Modern M6.5+ set: all 1,709 worldwide M6.5+ earthquakes in the USGS catalogue from 1990 to October 2026, reduced to **1,270 independent main shocks** by removing aftershocks (Gardner–Knopoff style windows). Sun, Moon and planet positions for every event came from the ephemeris tool in both the sidereal (Lahiri) and tropical frames. **Pre-registration.** The 23 rules, their thresholds, the corrected mandala lists and the interpretation rule were written down before any test (file preregistration_rules.md). Events before 2008 form the discovery set (632 events) and events from 2008 form the confirmatory holdout (638). Expected frequencies come from a 6-hourly background grid over 1990–2026. The Bonferroni threshold for 23 tests is 0.0022.

Rule (pre-registered)	Discovery p (before 2008)	Holdout p (2008+)	Holdout observed vs expected
R1 mandala [sid]	0.935	0.396	140/178/161/159 vs 142/189/165/141 (max ratio 1.12 (Varuna))
R2 Moon sharp group [sid]	0.631	0.039	111 vs 94.5 (1.17)
R8 Jupiter=Saturn nakshatra [sid]	0.761	0.614	20 vs 20.9 (0.95)
R11 PT season [sid]	0.440	0.173	108/120/93/119/110/88 vs 105/109/110/108/103/103 (max ratio 1.11 (1))
R1 mandala [trop]	0.824	0.151	142/214/151/131 vs 142/189/165/142 (max ratio 1.13 (Anala))
R2 Moon sharp group [trop]	0.501	0.585	93 vs 94.5 (0.98)
R8 Jupiter=Saturn nakshatra [trop]	0.267	0.893	16 vs 21.0 (0.76)
R11 PT season [trop]	0.535	0.599	95/120/99/114/107/103 vs 103/106/109/110/107/102 (max ratio 1.13 (1))
R3 Moon in Rikta tithi	0.973	0.193	137 vs 127.8 (1.07)
R4 within +/-1 tithi of new/full moon	0.972	0.650	122 vs 125.5 (0.97)
R5 eclipse within +/-15 d	0.260	0.309	177 vs 171.0 (1.04)
R6 eclipse within +/-7 d	0.304	0.343	113 vs 108.7 (1.04)
R7 Mars retrograde	0.776	0.344	63 vs 59.7 (1.06)
R7 Mercury retrograde	0.528	0.067	138 vs 122.4 (1.13)
R7 Jupiter retrograde	0.965	0.980	169 vs 192.0 (0.88)

Rule (pre-registered)	Discovery p (before 2008)	Holdout p (2008+)	Holdout observed vs expected
R7 Venus retrograde	0.739	0.846	41 vs 47.2 (0.87)
R7 Saturn retrograde	0.252	0.071	252 vs 233.6 (1.08)
R9 planet pair within 3 deg	0.253	0.327	134 vs 129.0 (1.04)
R10 Moon within 5 deg of planet/Sun	0.875	0.288	99 vs 93.6 (1.06)
R12 mandala vs mag [sid]	0.101	0.782	H=1.08
R12 mandala vs depth [sid]	0.684	0.914	H=0.52
R12 mandala vs mag [trop]	0.973	0.701	H=1.42
R12 mandala vs depth [trop]	0.672	0.804	H=0.99

Result. No rule passes the threshold on the holdout. The smallest holdout p-value is 0.039 (Moon in a sharp nakṣatra, sidereal frame, 111 vs 94.5 expected), and it points the opposite way in the discovery set (91 vs 93.6, p=0.63), so it is not replicated; the same rule in the tropical frame gives 93 vs 94.5. Mandalas are spread evenly (holdout 140/178/161/159 vs 142/189/165/141 expected, p=0.40), eclipse proximity is at chance (ratio 1.04), and neither magnitude nor depth differs between mandalas. In the original 15-event sample the sharp-group excess was 5 vs 2.2; with 638 independent events it shrinks to 1.17× and is not significant, a typical sign of a chance cluster in a small sample.

Power. At the corrected threshold, the 638 holdout events would reliably (80% of the time) detect an excess of roughly 35% in a rule like the sharp group, and would miss smaller effects such as the 17% seen. That is why section 12 and the Data-First Study repeat the tests on much larger samples.

6. The two four-circle systems: Parāśara Tantra and Brihat Saṃhitā

6.1 Parāśara Tantra Chapter 18

Verse 18.2 tells the founding story: flying, winged hills once shook the earth until Indra clipped their wings; he promised that henceforth earthquakes would come only from four causes, named for four elements/deities: **Anila** (wind or gas), **Anala** (fire), **Indra** (the king of the gods, here the benign mode) and **Varuṇa** (waters/ocean). Each day and each night is divided into four parts, and quakes in those parts are assigned to the four in that order. Eclipses and anomalous planetary movement are also said to induce earthquakes. The mandala of an actual quake is read from the Moon's nakṣatra at the time (18.3–18.6), optionally cross-checked against the time-of-day part.

Mandala	Nakṣatras (Moon in…)	Nature of the shaking (what the text says)	Body, society, regions	Our events
Anila (Vāyu: wind, gas) — 18.3	Svāti, Abhijit, Dhaniṣṭhā, Aśvinī, U.Phalgunī, Hasta, Citrā	Severe wind that destroys trees, flowers and plantations.	Delirium, respiratory disease, anxiety, fever. Doctors, traders, courtesans, sculptors, painters, poets and debaters suffer. Regions: Kuru, Paurava, Trigarta, Sauvīra, Yaudheya, Kṣudraka, Śibika.	Bhuj 2001 (Dhaniṣṭhā), İzmit 1999 (Svāti), Venezuela 2026 (Svāti) = 3
Anala (Agni: fire) — 18.4	Kṛttikā, Viśākhā, P.Bhādrapadā, Bharaṇī, P.Āṣāḍhā, Maghā, Puṣya, P.Phalgunī	Reduced rainfall, river flow and water in lakes and springs (a drying, heating mode).	Mouth, eye and skin diseases. Especially the south-east direction; Daśārṇa provinces troubled.	Tōhoku 2011 (Kṛttikā), Java 2006 (Kṛttikā), Sulawesi 2018 (Bharaṇī), Myanmar 2025 (P.Bhādrapadā) = 4
Indra — 18.5	Jyeṣṭhā, U.Āṣāḍhā, Śravaṇa, Rohiṇī, Mṛgaśirā, Anurādhā, Punarvasu	The favourable mode: good rains, abundant food and drink, religious festivals.	A famous clan chief is nevertheless foretold to be affected (scan partly unreadable). Regions: Kailāsa, Malla, Halava, Hala and neighbours.	Sumatra 2004 (Mṛgaśirā), Kashmir 2005 (Jyeṣṭhā), Bam 2003 (Śravaṇa), Nepal 2015 (Punarvasu) = 4
Varuṇa (waters) — 18.6	Śatabhiṣā, U.Bhādrapadā, Revatī, Ārdrā, Āśleṣā, Mūla	Continuous rain breaking fields, cities and trees (a flooding, wet mode).	Diarrhoea, hiccups, eye disease. Regions: Kirāta, Kāśmīra, Aparānta, Kaukureya, Sauryārika, Cedi, Vatsa, Saindhava, and coastal and riverbank peoples.	Haiti 2010 (Mūla), Sichuan 2008 (Āśleṣā), Turkey–Syria 2023 (Āśleṣā), Morocco 2023 (Ārdrā) = 4

Correction from the Sanskrit. Checked on the page images, the printed English translation differs from the Sanskrit in three places: 18.3 reads Āryamṇa (U.Phalgunī), not P.Phalgunī; 18.4 reads Bhaga (P.Phalgunī) and Āpa (P.Āṣāḍhā), where the English gives U.Phalgunī and omits P.Āṣāḍhā; 18.5 reads Āditya (Punarvasu), where the English gives Hasta. The Varuṇa line confirms Ārdrā (Rudra). With these, the 27 nakṣatras divide 6 + 8 + 7 + 6 (Abhijit is extra, under Anila). None of the 15 main events had the Moon in a nakṣatra affected by the corrections, so their assignments are unchanged; all later tests use the corrected lists.

How the mandalas were applied

By Moon nakṣatra (primary): each quake's Moon nakṣatra gives its mandala (table above). **By time of day** (secondary): local solar hour gives one of four parts in the order Anila, Anala, Indra, Varuṇa; the nakṣatra and time-of-day mandalas agreed in only 3 of 15 events (about 3.75 expected by chance), so the two readings are not mutually consistent. **Result:** the events spread 3/4/4/4 across the four mandalas, essentially uniform. The text's predicted effects (wind damage, drought, good rains, flooding) could not be checked quantitatively, because the quakes were chosen for casualties, not for the weather or disease around them.

Other mandala systems used in the study

System	What it is	Meaning in the study
Brihat Saṃhitā ch.32 (four circles)	Varāhamihira's four-circle model, which also uses Vāyu, Agni, Indra and Varuṇa but assigns nakṣatras to the circles differently from Parāśara Tantra.	Recorded per event as a second, independent classification: Vāyu 4, Varuṇa 4, Indra 3, Agni 4 (uniform; no signal). The original chapter was later supplied and read from the Sanskrit; all four mandalas and the time-of-day rule are tested in this section.
Venus's six circles (PT Ch.9)	Nakṣatra blocks starting at Bharaṇī: Ādya, Rohita, Dāruṇa, Virocana (four each), Ūrdhvadaṇḍa and Tīkṣṇa (five each). The first four are also called Vāta.	Used to classify where Venus stood at each quake; every circle within chance in both sidereal and tropical frames.
Moon nakṣatra groups (PT Ch.14)	Dhruva (fixed), Mṛdu (soft), Ugra (fierce), Tīkṣṇa (sharp), Kṣipra (swift), Cara (moving), Miśra (mixed).	Used for the Ch.14 test; the sharp group (Ārdrā, Āśleṣā, Jyeṣṭhā, Mūla) had 5 events (p=0.059, not significant).

6.2 Brihat Saṃhitā Chapter 32, read from the Sanskrit

Source. Varāhamihira's Bṛhatsaṃhitā with Bhaṭṭotpala's commentary and Hindi translation (Chowkhamba Vidyabhawan), chapter 32, Bhūkampalakṣaṇādhyāya, printed pp. 410–419, read from the page images. **Content.** Verses 1–2 list rival causes (large water creatures, tired world-elephants, colliding winds, unseen merit and demerit). Verses 3–7 divide day and night into quarters. Verses 8–22 give four mandalas of seven nakṣatras each, with omens that appear **seven days before** the shock. Verse 23: an earthquake's result arrives within six months, a nirghāta (thunderous portent) within two.

Mandala (BS)	Nakṣatras (Moon in…)	Time of day (Gārgya, 3–7)	Omens and effects (summary)
Vāyavya (8–11)	U.Phalgunī, Hasta, Citrā, Svāti, Punarvasu, Mṛgaśirā, Aśvinī	First half of the day	Smoke-filled directions, dust, broken trees, dim sun. Crop, water and herb loss; swelling, asthma, madness, fever, cough; traders, courtesans, soldiers, physicians, singers, craftsmen; Saurāṣṭra, Kuru, Magadha, Daśārṇa, Matsya.
Āgneya (12–15)	Puṣya, Kṛttikā, Viśākhā, Bharaṇī, Maghā, P.Bhādrapadā, P.Phalgunī	Second half of the day	Meteors, burning skies and fire carried by wind. Clouds and tanks dry up; hatred among kings; skin disease, fever, jaundice; Aśmaka, Aṅga, Bāhlīka, Kaliṅga, Vaṅga, Draviḍa, Śabara.
Aindra (16–19)	Abhijit, Śravaṇa, Dhaniṣṭhā, Rohiṇī, Jyeṣṭhā, U.Āṣāḍhā, Anurādhā	First half of the night	Dark, thundering, lightning-bearing clouds shower rain. Famous families, kings and chiefs harmed; diarrhoea, throat and mouth disease; Kāśī, Gandhāra, Pauravas, Kirāta, Abhisāra, Mālava, Surāṣṭra.
Vāruṇa (20–22)	Revatī, P.Āṣāḍhā, Ārdrā, Āśleṣā, Mūla, U.Bhādrapadā, Śatabhiṣā	Second half of the night	Blue-lotus, bee-coloured clouds, heavy rain. Those living by sea and rivers destroyed; excess rain; Gonarda, Cedi, Kukura, Kirāta, Videha.

Differences from Parāśara Tantra. Both use four deities, but the lists differ: The BS places Mṛgaśirā and Punarvasu in Vāyu (PT: Indra), Dhaniṣṭhā in Indra (PT: Anila) and P.Āṣāḍhā in Varuṇa (PT: Anala); the rest agree (for example Ārdrā, Āśleṣā and Mūla in Varuṇa, and U.Phalgunī in Vāyu/Anila). BS gives seven each and Abhijit sits in Indra. **Tests.** (i) The 4-class mandala of the Moon's nakṣatra on the Modern M6.5+ set (1,270 events): sidereal holdout 171/158/130/179 vs 165/166/142/165 expected (p=0.44; tropical p=0.13). (ii) The time-of-day rule, with local solar time split into 06–12, 12–18, 18–24 and 00–06: the nakṣatra mandala matched the time mandala in 152 holdout events vs 159.2 expected (p=0.76). The Vāyu list alone gave 171 vs 165 (p=0.32). Nothing passes the threshold for three tests (0.0167). **Not testable.** The omens (smoke, meteors, cloud colours) and regional effects are not recorded for these events, and the six-month result rule would need outcome data.

Vāyu rule. Moon in Aśvinī, Mṛgaśirā, Punarvasu, U.Phalgunī, Hasta, Citrā or Svāti: holdout 171 vs 165 expected (p=0.32); combined with local solar time 03:00–09:00 (last quarter of night, first of day): 43 vs 41

(p=0.42).

6.3 Contrast of the two texts

Both texts describe four earthquake types named after the same four powers and read from the Moon's nakṣatra, so they look alike, but they differ in structure, lists, timing rules and detail. The tests show they also give the same answer: no signal.

Feature	Parāśara Tantra (Ch.18, Iyengar ed.)	Brihat Saṃhitā (Ch.32, Varāhamihira)
Four types	Anila, Anala, Indra, Varuṇa	Vāyavya, Āgneya, Aindra, Vāruṇa (the same four)
Origin story	Indra clipped the wings of flying hills and limited quakes to four causes (18.2)	Opens with rival causes: water creatures, tired world-elephants, colliding winds, unseen merit and demerit (1–2); then the four types
Nakṣatras per type	6 / 8 / 7 / 6 (Abhijit extra, in Anila), after correcting the Sanskrit	7 / 7 / 7 / 7 (Abhijit in Aindra)
Agreement	Of the 27 nakṣatras, 23 fall in the corresponding type in both texts. The four that differ: Mṛgaśirā and Punarvasu (PT Indra, BS Vāyu), Dhaniṣṭhā (PT Anila, BS Indra), P.Āṣāḍhā (PT Anala, BS Varuṇa); Abhijit also differs.	
Time of day	Day and night each in four parts, in the order Anila, Anala, Indra, Varuṇa	Gārgya: first half of day Vāyu, second half Agni, first half of night Indra, second half Varuṇa; Parāśara's quarters quoted too (3–7)
Portents	Effects described by type; also eclipses and planetary anomalies as inducers	Omens listed per type and said to appear seven days before the quake; result within six months, a nirghāta within two (23)
Effects	Wind and plants, drought, good rain, flooding; named diseases, trades and peoples	Same themes with fuller lists (smoke and dust, meteors and fire, thunder clouds, blue-lotus clouds), plus regions
Other chapters	Rāhu, comets, planets, tithi, nakṣatra groups (Ch.3–16, 19) add testable rules	Ch.32 alone; a scheme of its own

Test	Parāśara Tantra	Brihat Saṃhitā
15 deadly events: spread over four types	3/4/4/4 (Anila/Anala/Indra/Varuṇa)	Vāyu 4, Agni 4, Indra 3, Varuṇa 4
1,270 events, holdout: Moon-nakṣatra type vs expected	140/178/161/159 vs 142/189/165/141 (p=0.40 sidereal; 0.15 tropical)	171/158/130/179 vs 165/166/142/165 (p=0.44 sidereal; 0.13 tropical)
Time-of-day agreement	3 of 15 vs 3.75 expected (15 events only; not rerun on the large set)	152 vs 159.2 expected on the holdout (p=0.76)
Severity (magnitude, depth) by type	No difference (Kruskal–Wallis, all p>0.10)	Not run (same Moon positions give near-identical groups)
Verdict	Not supported	Not supported

Reading the comparison. (1) The two systems are not independent tests: for the Moon positions of the 1,709 events they give the corresponding type in 85.7% of cases, against 85.2% for random times, so the same underlying nakṣatra drives both. A pass or a fail in one says almost the same as in the other. (2) The differences are in four nakṣatras (about 15% of the sky), which is too few for the tests to separate the two lists; neither list is closer to the data. (3) What differs most is the detail: Parāśara Tantra adds eclipses, comets, tithi and planetary rules that could be tested, while Brihat Saṃhitā adds omens and regional effects that could not, because no event records them. (4) The corrected Parāśara Tantra lists (section 6.1) were used throughout, so the comparison is between the Sanskrit readings of both texts, not the printed English.

6.4 Cross-check against the English translation

Why this check. The first reading of the Brihat Saṃhitā came from page images of the Sanskrit, with no automatic text recognition for Devanagari, so a misread letter was possible. The printed English translation, supplied afterwards,; it is an independent reading of the same verses, so wherever the two agree the risk of a misreading falls sharply, and where they differ we can see which version matters. **What we found.** The English matches the earlier Sanskrit reading for three of the four circles exactly (Vāyu, Agni, Indra), for the six-month result window (32.23) and for the repeat-quake days 3, 4, 7, 15, 30 and 45 (32.34), which also confirms that the '15' reading was right. There is **one disagreement**: the English prints U. Āṣāḍhā in the Varuṇa circle (32.20), but it already placed it in the Indra circle (32.16) and leaves out U. Bhādrapadā, which the Sanskrit gives to Varuṇa. This looks like a printing slip in the English, so the Sanskrit list stays the main reading, and the English-as-printed lists were run as variants. None of the Deadly 15 has the Moon in either of those two stars, so for them the result is identical under every version. **New tests from the English.** It also states things we had not tested: that the time-of-day circle overrides the star circle (32.27), what happens when the two disagree (32.28–31), how far each type of quake is felt (200, 180, 160 and 110 yojanas for

Vāyu, Varuṇa, Indra and Agni; 32.33), and that an earthquake within seven days after an eclipse ends is a bad sign (5.92). All were written down in advance (file preregistration_bs_english_crosscheck.md) and run on every data set: the 9,368 earthquakes of 1990–2026 (and its before/after-2008 halves), 6,111 events of 1964–1989 and 670 of 1900–1963. **Result.** Whichever version of the lists is used, there is no signal: the four-circle counts sit within 2–3% of chance (smallest p=0.04 before correction, bar 0.005), and the time-of-day rule, the felt-distance ordering (correlation with magnitude about 0.00) and the post-eclipse window all look like chance. One of the 50 tests, the 16 pairings of star circle and time circle in 1964–1989, reached p=0.0048, but it failed in 1990–2026 (p=0.50 in all years, 0.67 in the 2008+ holdout) and in 1900–1963 (p=0.65), so under the rule we set it is not supported. Fifty tests at a 0.005 bar produce about one such result in four runs by luck alone.

Verse (BS)	Sanskrit reading (earlier)	English translation (supplied)	Agreement
32.7 time of day	Vāyu, Agni in the two halves of the day; Indra, Varuṇa in the two halves of the night	'First, second, third and fourth six hours of day and night'	Consistent (four six-hour quarters)
32.8 Vāyu circle	U.Phalgunī, Hasta, Citrā, Svāti, Punarvasu, Mṛgaśirā, Aśvinī	Same seven	Match
32.12 Agni circle	Puṣya, Kṛttikā, Viśākhā, Bharaṇī, Maghā, P.Bhādrapadā, P.Phalgunī	Same seven	Match
32.16 Indra circle	Abhijit, Śravaṇa, Dhaniṣṭhā, Rohiṇī, Jyeṣṭhā, U.Āṣāḍhā, Anurādhā	Same seven	Match
32.20 Varuṇa circle	Revatī, P.Āṣāḍhā, Ārdrā, Āśleṣā, Mūla, U.Bhādrapadā, Śatabhiṣā	Same, but U.Āṣāḍhā printed where U.Bhādrapadā belongs	**One difference** (apparent printing slip)
32.23 timing	Earthquake results within six months	Same	Match
32.34 repeat days	Second quake on day 3, 4, 7, 15, 30 or 45 (tested with 14 and 15)	Same; day 15	Match; '15' confirmed
32.27–31	Not tested before	Time circle overrides star circle; mixed-circle outcomes	New test
32.33	Not tested before	Felt distance 200, 180, 160, 110 yojanas (Vāyu, Varuṇa, Indra, Agni)	New test
5.92	Not tested before	Earthquake within 7 days after an eclipse: rulers die	New test

Re-run results. Each cell shows observed ÷ expected (1.00 = chance) and the p-value before correction; the bar is 0.005. The columns are the Modern M5.5+ set (all years and the 2008+ holdout), Older set 1 and Older set 2.

Test (frozen in advance)	1990–2026 all years	2008–2026 holdout	1964–1989	1900–1963
1 mandala, Sanskrit lists	1.02× , p=0.119	1.02× , p=0.719	1.02× , p=0.682	1.11× , p=0.162
2 mandala, English printed (U.Ashadha in Indra)	1.03× , p=0.086	1.02× , p=0.503	1.02× , p=0.675	1.11× , p=0.171
3 mandala, English printed (U.Ashadha in Varuna)	1.03× , p=0.075	1.02× , p=0.630	1.01× , p=0.690	1.11× , p=0.282
4 mandala, both U.Ashadha and U.Bhadrapada set aside	1.02× , p=0.075	1.02× , p=0.493	1.01× , p=0.734	1.10× , p=0.268
5 time-of-day circle (also v.27 precedence)	1.01× , p=0.616	1.02× , p=0.818	1.02× , p=0.821	1.05× , p=0.581
6 nakshatra circle = time circle	0.98× , p=0.828	0.99× , p=0.701	1.00× , p=0.579	0.94× , p=0.843
7 all 16 circle pairs (v.28-31)	1.08× , p=0.504	1.12× , p=0.674	1.15× , p=0.005	1.21× , p=0.654
8 felt distance order by nakshatra circle	p=0.638	p=0.297	p=0.813	p=0.765
9 felt distance order by time circle	p=0.314	p=0.288	p=0.283	p=0.799
10 eclipse within 7 days before the earthquake	1.01× , p=0.394	1.06× , p=0.114	1.00× , p=0.537	0.90× , p=0.819

Takeaway. A second, independent reading of the verses closes most of the worry about misreading the Sanskrit. The two readings agree almost everywhere, the one difference is a likely printing slip that makes no difference to the results, and the newly testable statements show no link to earthquakes either.

7. The crescent-Moon 'carriage pole' omen (Brihat Saṃhitā 4.12; Parāśara Tantra Ch.4)

First version (proxy). The Parāśara Tantra describes a 'yoke' crescent as an earthquake portent. As a first test the crescent's tilt was approximated by the Moon's ecliptic latitude (main term 5.128° × sin F): north tip

raised, among waxing-crescent events 12–48° from the Sun. North tip raised (Moon ecliptic latitude >0, main term 5.128° sin F) among waxing-crescent events (12–48° from the Sun): holdout 42 of 67 vs 33.6 (p=0.026), discovery 40 of 74 vs 37.1 (p=0.29) ; the family threshold of 4 tests is 0.0125, so it is not supported.

Second version (true horizon). What the verse says. Brihat Saṃhitā 4.12 (English translation supplied): if the new Moon 'appears stretched from North to South presenting the appearance of a carriage pole', there will be an earthquake within that month. A footnote adds that the Moon is to be watched on the first or second day after the new moon. The proxy above only approximates the tilt, so we computed it as an observer would see it. **What we did.** For each earthquake we found its lunar month, then the first evening on which a thin crescent is really visible (Sun 3° below the horizon, Moon at least 3° up and at least 8° from the Sun), and measured how level the line joining the two horns is, and whether the Moon stands near due west so that this line runs north to south. A 'level yoke month' means a tilt of 20° or less (and the Moon within 30° of west). The verse does not say where the observer stands, so we tried two readings, fixed in advance (file preregistration_bs_crescent_true_horizon.md): an observer at Ujjain, the traditional meridian of the text, and an observer at the epicentre. The Moon's latitude calculation was first checked against the textbook's worked example (matched to 0.00001°). **Result.** A level yoke appears in about 40% of months (the Moon is usually in the west at dusk in these latitudes), and earthquakes fall in such months no more often than chance in any era: 1.03× for 1990–2026 (Ujjain reading), 0.97× for 1964–1989 and 0.95× for 1900–1963. Under the epicentre reading the 2008–2026 holdout shows 1.04× (p=0.019, bar 0.0125), but 1990–2007, 1964–1989 and 1900–1963 show 1.00×, 1.00× and 1.02×, so it is not supported under the rule we set. (The epicentre p-values are likely generous, because quakes in the same month are counted separately.)

Test (frozen in advance)	1990–2026	2008+ holdout	1990–2007	1964–1989	1900–1963
U yoke (Ujjain, date-shift null)	1.03×, p=0.435	1.08×, p=0.425	0.99×, p=0.641	0.97×, p=0.858	0.95×, p=0.659
L yoke (epicentre, binomial)	1.01×, p=0.186	1.04×, p=0.019	1.00×, p=0.425	1.00×, p=0.599	1.02×, p=0.377
L level (epicentre, binomial)	1.01×, p=0.186	1.04×, p=0.016	1.00×, p=0.421	0.99×, p=0.646	1.01×, p=0.398

Takeaway. Even when the Moon's shape is worked out exactly as a person under the sky would see it, the 'carriage pole' Moon does not come before more earthquakes than chance. The columns are the Modern M5.5+ set (all years, 2008+ and 1990–2007) and Older sets 1 and 2.

8. Regional tests: do the rules work in particular earthquake belts?

Why this check. A single worldwide pool could hide an effect that exists only in one earthquake belt, and the old texts themselves tie effects to regions. We therefore split all earthquakes into six fixed belts (Japan–Kurils–Korea, Philippines–Taiwan, Indonesia, China, Himalaya–India–Myanmar, and West and Central Asia including Turkey, Iran and the Hindu Kush) and ran seven rules in each: the Brihat Saṃhitā circle, the Parāśara mandala, the Moon near the Sun or a planet, the Moon in the sharp star group, the Sun in Viśākhā, and the crescent test seen from the epicentre and from the middle of the belt. That is 42 tests per period, so the bar for a claim is very strict (p<0.0012). The boxes and rules were fixed in advance (file preregistration_regional_tests.md). **Result.** No belt-and-rule pair passes. The smallest holdout p-value is 0.004 (the region-centre crescent for the Himalaya belt), and it fails in 1964–1989 (p=0.33). Among 42 holdout tests, 3 reach p<0.05 against about 2 expected from chance alone, and the same happens in the other periods. A fair warning: some belts are small (China has only 71 events in 1990–2026), so for them only large effects could be seen. **The one pattern worth noting.** The 'level yoke' crescent (Brihat Saṃhitā 4.12) seen from the epicentre appears slightly more often in the month of a quake in two belts, in every period. For Japan–Kurils–Korea the ratio to chance is 1.18, 1.18, 1.24 and 1.10 in the four periods; for Indonesia it is 1.07, 1.01, 1.06 and 1.23. Pooled over all periods, Japan gives 252 yoke months against 211 expected (1.19×, p=0.0014) and Indonesia gives 1.05× (p=0.002). With a date-shift check that allows for clustering in time, Japan's p rises to 0.03 and Indonesia's to 0.07. Because the region and the pooling were chosen after looking, and 42 tests were run, these pooled values are 0.06 and 0.09 after correction, so they are **not significant**. We record them as a faint descriptive pattern, not a finding. All earthquake data from 1900 to 2026 has now been used, so a fair confirmation needs new earthquakes after October 2026 (about 18 qualifying Japan-belt events a year, so several years of waiting) or a pre-1900 catalogue. **Data.** The Modern M5.5+ set (split at 2008), Older set 1 and Older set 2.

Region (fixed box)	Events: 1990–2007 / 2008–2026 / 1964–89 / 1900–63	Smallest p in the 2008+ holdout (of 7 rules)	Rules with p<0.05: holdout / 1964–89 (of 7)
Japan-Kurils-Korea	341 / 310 / 516 / 80	0.0519 (R7)	0 / 2
Philippines-Taiwan	188 / 190 / 284 / 49	0.3010 (R1)	0 / 1
Indonesia	602 / 544 / 833 / 72	0.0266 (R5)	1 / 1

Region (fixed box)	Events: 1990–2007 / 2008–2026 / 1964–89 / 1900–63	Smallest p in the 2008+ holdout (of 7 rules)	Rules with p<0.05: holdout / 1964–89 (of 7)
China	33 / 38 / 67 / 0	0.1923 (R4)	0 / 1
Himalaya-India-Myanmar	137 / 110 / 152 / 21	0.0040 (R7)	2 / 0
West-Central Asia	147 / 148 / 204 / 33	0.3118 (R1)	0 / 1

Period	Japan–Kurils–Korea: events	Level-yoke months observed vs expected	Ratio	p (binomial)
1990-2007 discovery	341	65 vs 55.0	1.18×	0.083
2008-2026 holdout	310	73 vs 62.0	1.18×	0.070
Era A 1964-1989	516	96 vs 77.7	1.24×	0.016
Era B 1900-1963 (M7+)	80	18 vs 16.3	1.10×	0.364
All periods pooled	1247	252 vs 211.0	1.19×	0.0014 (date-shift 0.030; 0.058 after correction for 42 tests)

Takeaway. Looking belt by belt does not rescue the rules. The Himalaya, China, West and Central Asia and Philippines belts show nothing. A weak, consistent hint appears for the Moon-shape rule in Japan and Indonesia, but it is not strong enough to count after allowing for how many comparisons were made, and it can only be tested properly on future earthquakes.

9. The TG-SAI model

9.1 Findings about the code

• Each feature is a 7-day peak. Because cos(2Δ) peaks at both new and full moon, nearly every 7-day window contains a syzygy, so the syzygy feature is close to 1 for most events (instantaneous vs. peak correlation only 0.15). It carries little information. • Saturn, Jupiter and Mars “alignments” use crude mean-longitude formulas, not true geocentric positions. • Training positives and negative samples are generated asymmetrically, which can inflate apparent skill. • Tithi is the discrete form of the syzygy feature (Δ/12°); five rule-derived features (tithi, karaṇa, Moon-nakṣatra group, Moon–planet and planet–planet separation) could be added, but 15 events cannot train them.

9.2 Retrained properly

Fixes applied: (a) matched negatives, i.e. random times with latitudes drawn from the real events, so positives and negatives are generated the same way; (b) instantaneous features instead of 7-day peaks (which sit near 1 by construction); (c) true geocentric planet longitudes from the ephemeris instead of the crude mean-longitude formulas; (d) evaluation by 5-fold cross-validation, by a time split (train before 2008, test after) and against 30 shuffled-label runs. 1,270 positives and 1,270 negatives, RandomForest.

Model	Features	AUC, 5-fold CV	AUC, time split	AUC, shuffled labels (mean / max)
A: repository formulas, instantaneous	7	0.501	0.505	0.492 / 0.513
B: true geocentric planets	7	0.505	0.489	0.489 / 0.515
C: B + Vedic features (Moon, tithi, mandala)	15	0.514	0.499	0.495 / 0.524

An AUC of 0.50 is chance. The model cannot tell a real M6.5+ earthquake time from a random time, with or without the Vedic features, so the high probabilities reported by the original code came from its sampling design, not from skill.

9.3 Scored on older data it never saw

The model was scored once under frozen rules (file preregistration_tgsai_bottomup_historical.md). We trained the corrected model once on the Modern M5.5+ set against matched 'non-earthquake' moments, then scored it, without any retraining, on earthquakes it had never seen. In plain terms, AUC is the chance that the model ranks a real earthquake moment above a random one: 0.50 is a coin toss and 1.00 is perfect. The repository's own formulas (Model A) scored 0.503 on 1964–1989 (95% range 0.492–0.512), 0.502 on 1900–1963, and 0.509 on the repository's own 1850–1950 file. The two improved versions (true planet positions; plus the Vedic star, tithi and mandala features) were equally flat, between 0.499 and 0.514. The pass mark was 0.52 with the lower end of the range above 0.50; none came close. The 1850–1950 file gave one nominal p-value of 0.035 for Model A, but with nine model-by-dataset comparisons that is within chance and the AUC itself is only 0.509.

Test (frozen in advance)	Data scored	Result	Verdict
TG-SAI Model A (repository formulas), trained on 1990–2026	1964–1989, 6,111 events	AUC 0.503 (0.492–0.512)	No skill (bar: 0.52)
TG-SAI Models B and C	1964–1989	AUC 0.499 and 0.501	No skill
TG-SAI Models A, B, C	1900–1963, 670 events	AUC 0.502, 0.509, 0.514 (ranges include 0.50)	No skill
TG-SAI Models A, B, C	Repository file 1850–1950, 5,003 events	AUC 0.509, 0.505, 0.507	No skill (one nominal p=0.035, not significant after 9 comparisons)

9.4 The repository's own training file

The file. The repository's notebooks folder holds Inputs_Mega_Quake_Topocentric_Analysis.csv (file tgsai_training_file_Inputs_Mega_Quake_Topocentric_Analysis.csv). It has 104 great earthquakes of magnitude 8.0 to 9.5, from 1700 to 2025 (100 from 1900 onward), each with eleven sky columns: Moon declination, a 'lunar torque index', the lunar node, the Moon–Sun phase angle, the Moon's distance, and sidereal longitudes of the Moon, Sun, Mars, Jupiter and Saturn. It contains only earthquakes, with no non-earthquake examples, so the repository's original training run cannot be reproduced exactly from it; we can audit it and test it.

Audit. (1) Moon distance is constant (384,467 km in all 104 rows), so that column carries no information; the repository's 'perigee' measure cannot be computed from it. (2) The 'lunar torque index' is simply the absolute Moon declination divided by 10 (maximum difference 0.00005), not a separate quantity. (3) The Sun's and Moon's longitudes are sound (Sun within 0.03° of our ephemeris; Moon median difference 0.7°, as expected for a topocentric position). (4) The planet columns are not true positions: against our validated ephemeris the median differences are 25° for Mars, 7° for Jupiter and 5° for Saturn (up to 52°), and they match neither the true planets nor the repository's crude formulas exactly. This confirms, on the training data itself, the 'crude planet formula' problem found in the code. (5) The node column is the tropical mean node, although the other longitudes are sidereal. (6) The earliest rows (1700, 1843) fall outside the validity of our planet elements and are dropped from our own tests.

Tests. These were run after receiving the file, not pre-registered, so they are exploratory; with six tests the bar is 0.05 ÷ 6 = 0.0083. Expected counts come from random times spread uniformly over 1900–2025 (200,000 sky positions).

Test (100 great earthquakes, 1900–2025)	Observed vs expected	p-value
Moon in the sharp group	12 vs 14.9 (0.80×)	0.83
Sun in Viśākhā	7 vs 3.6 (1.93×)	0.072
Moon within 5° of the Sun or a planet	13 vs 14.6 (0.89×)	0.72
Moon within 12° of a new or full moon	16 vs 13.1 (1.22×)	0.24
PT mandala by Moon nakṣatra (four classes)	27/29/28/16 vs 22.3/29.4/25.9/22.4	0.39
BS mandala by Moon nakṣatra (four classes)	34/27/21/18 vs 26.0/25.7/22.2/26.1	0.16

Fixes applied, then the model recomputed. Every defect found in the audit was corrected before refitting: planet longitudes replaced by true geocentric positions, the node converted to the sidereal frame (tropical node minus the Lahiri ayanāṃśa), the constant Moon distance replaced by a computed Earth–Moon distance (so perigee is real), and the 'torque' column kept only as the declination it is. The 102 great earthquakes from 1850 onward were set against 2,040 random times in the same years (matched negatives, 10 repeats with new negatives each time), with a RandomForest, 5-fold cross-validation, a time split (train before 1970, test from 1970) and shuffled labels as the chance reference.

Model on the supplied file	AUC, 5-fold CV (mean; range)	AUC, time split	AUC, shuffled labels (mean / max)
Fixed features (true planets, sidereal node, true Moon distance)	0.540 (0.494–0.569)	0.42	0.492 / 0.538
Same, node left tropical as in the file	0.526 (0.487–0.563)	0.42	0.487 / 0.528
Fixed features plus Vedic features (mandalas, sharp group, tithi)	0.547 (0.497–0.581)	0.44	0.484 / 0.544

Re-scoring the Deadly 15. The corrected model was trained on the 100 great earthquakes with the two that also belong to the Deadly 15 (Sumatra 2004, Tōhoku 2011) removed, then used to score the Deadly 15 against 3,000 random times, repeated 10 times. The AUC is 0.550 (range 0.508–0.601). Eight of the 15 score above the median random time and four above the 75th percentile (Java 2006, Sumatra 2004, Myanmar 2025, İzmit 1999); Turkey–Syria 2023 (8th percentile) and Sichuan 2008 (26th) score low. With only 15 events an AUC of 0.55 cannot be told apart from 0.50 (its 95% range is roughly 0.38 to 0.72), so the original table in

section 2 should be read as the repository's uncorrected output, and the corrected model gives no better than chance (file tgsai_deadly15_rescore.csv).

Reading. Fixing the frame changes the node result by about 0.01, which is within the spread between repeats, so the frame error did not matter. After all fixes the cross-validated AUC is 0.54–0.55, only just above the shuffled-label maximum (0.54), and it drops to 0.42–0.44 when the model must predict later earthquakes from earlier ones (below chance, so the small cross-validated edge does not generalise). With only 102 positives, that edge is not distinguishable from chance. The Sun in Viśākhā again leans high in this small sample (7 events, p=0.072, not significant), the same star window that was a chance lead before; nothing else stands out. The file does not rescue the model or any rule. It removes the 'file missing' uncertainty: it is a small, positives-only set with a constant column and approximate planet positions, and the corrected model on it performs at chance.

10. Aftershock recurrence (verse 18.7) and comets (verse 11.19)

10.1 Verse 18.7 on the Deadly 15

Data. The USGS ComCat catalogue for each Deadly 15 event: all quakes M4.5+ within 300 km, from 60 days before to 90 days after the main shock (4,022 rows; per-event row counts matched USGS's own count endpoint where it responded). **Rule tested.** Verse 18.7 says repeated tremors at intervals of 3, 4 or 7 nights, a fortnight, a month and a month and a half portend the fall of an important king. **Method.** For each event we counted how many of the six lags (3, 4, 7, 14, 30, 45 days) had at least one aftershock within ±12 h of the main shock, and separately how many gaps between successive aftershocks matched a lag. The baseline was ordinary aftershock decay (Omori's law, fitted per event), simulated 4,000 times with the same number of aftershocks, so the verse's intervals had to beat what plain decay produces. The magnitude cut of M5.0 was fixed as the primary before looking; M4.5 and M5.5 are sensitivity checks.

Event	Aftershocks M≥5.0 (0–90 d)	Foreshocks M≥5.0 (60 d)	Verse lags hit (observed)	Expected from ordinary decay
Sumatra 2004	132	2	5 of 6	4.7
Haiti 2010	17	0	0 of 6	0.7
Sichuan 2008	56	0	3 of 6	2.7
Kashmir 2005	46	1	1 of 6	2.3
Turkey-Syria 2023	39	0	3 of 6	2.3
Bam 2003	2	1	0 of 6	0.2
Bhuj 2001	9	0	1 of 6	0.8
Tohoku 2011	599	32	5 of 6	5.7
Izmit 1999	12	1	2 of 6	1.2
Nepal 2015	31	1	0 of 6	1.8
Venezuela 2026	1	1	too few	–
Java 2006	34	1	0 of 6	2.4
Sulawesi 2018	21	3	2 of 6	1.0
Myanmar 2025	4	2	0 of 6	0.4
Morocco 2023	0	0	too few	–

Pooled result. Lag hits: M5.0 → 22 observed vs 26.2 expected (13 events); M4.5 → 46 vs 50.2 (15 events); M5.5 → 7 vs 11.9 (11 events). Successive-gap hits: 14 vs 16.5, 21 vs 23.6, 11 vs 8.8. The observed hits are at or below what ordinary aftershock decay gives, so the verse's recurrence intervals show no excess. **Foreshocks:** the large counts are Tōhoku (32 M5+, including the M7.3 two days earlier) and Sulawesi 2018 (3 M5+, including the M6.1 foreshock hours before); most other events had 0–2. **Caveats:** Venezuela and Morocco have almost no aftershocks at this magnitude, Bam and Myanmar very few, so those rows say little; Java 2006's decay fit hit the lower bound; early-sequence catalogue completeness is poor (especially Sumatra 2004 and Tōhoku); the lag windows are our operationalisation of an ambiguous verse.

10.2 Completeness-corrected rerun and the larger study

Completeness-corrected rerun. Excluding the first 12 hours of each sequence (when small aftershocks are missing from catalogues), fitting the decay on the remaining window and pooling all events: lag hits 22 observed vs 24.4 expected at M5.0 (10 events; p=0.86 for an excess) and 46 vs 48.1 at M4.5 (14 events; p=0.79). Successive-gap hits: 12 vs 16.3 and 21 vs 24.3. The earlier conclusion is unchanged: no excess over ordinary aftershock decay.

Larger study. USGS M5.0+ aftershocks within 100–300 km and 0–90 days for 481 main shocks of M7.0+ (the Aftershock set) (11,608 records, 7,323 usable at M5.0). Omori-law baselines were fitted per sequence and simulated 150 times. Variants: tolerance 0.25, 0.5 and 1 day; lag sets {3,4,7,14,30,45}, {3,4,7,15,30,45}, {3,4,7}, {14,30,45}; lags from the main shock and gaps within a sequence; M5.0 and M5.5. In 24 variants no p-value was below 0.46; observed hits were at or below expectation almost everywhere (e.g. 454 vs 516 expected at M5.0, ±0.5 d, six lags). Both readings of “repeatedly” are therefore null.

10.3 Comets (verse 11.19)

An earthquake is said to precede a comet. Using the seven great comets of 1990–2026 (Hyakutake, Hale–Bopp, McNaught, Lovejoy, NEOWISE, Tsuchinshan–ATLAS, G3 ATLAS; perihelion dates from Wikipedia's Great comet page), a great-comet perihelion followed a quake within 12 months for 203 of 1,236 events vs 214.4 expected (p=0.81), and within 24 months for 384 of 1,205 vs 379.8 expected (p=0.41). No signal, though with only seven comets this test has little power.

Not testable. Chapter 19 haloes and sky brightening have no structured record per event, so that claim cannot be tested.

11. Sensitivity to the ayanāṃśa

Source. S. Madhavan, Muddle of Ayanamsa, read for its argument and its worked ayanāṃśa figures (the sections on precession and on the competing systems; the astronomical introduction in chapters 2–6 was not read in detail). The book's point is that the sidereal zodiac has no single agreed starting point. Systems differ in two things: the year the sidereal and tropical zodiacs coincided (Lahiri 285 AD, Newcomb 291 AD, Raman about 398 AD, traditional/Sepharial 498 AD, Swamikannu Pillai 532 AD, Burgess 560 AD) and, slightly, the rate of precession (about 50.3 seconds a year). On 5 January 1971 the book gives Raman 22°0′24″, Krishnamurti 23°21′, Lahiri 23°27′, traditional (Sepharial) 21°36′ and Swamikannu Pillai 23°28′27″, and Chidambara Iyer 20°24′15″ for 1983. The book also argues for the observation-based (Dṛg) method over the Vākya almanac, which is what the ephemeris tool computes.

Why it matters here. The Moon moves about 13.2° a day and a nakṣatra is 13.33° wide, so an ayanāṃśa shift of d degrees moves about d/13.3 of all events into a different nakṣatra: roughly 1 in 9 for Raman (1.4° below Lahiri), 1 in 7 for the traditional 498 AD value (1.9°), and almost every event for the tropical frame (23.85°, nearly two nakṣatras). Every nakṣatra number in this report (mandalas, groups, the 3/4/4/4 split) therefore depends on the ayanāṃśa chosen. The main analysis used Lahiri; the tropical frame was the only alternative shown. To reconcile the numbers we reran the large-sample mandala and sharp-group tests under each named system on the Modern M6.5+ set and over a sweep of 29 offsets from 4° below to 3° above Lahiri.

System (offset vs Lahiri)	15-event mandala counts (Anila/Anala/Indra/Varuṇa, 14 matched events)	Holdout p: PT mandalas	Holdout p: BS mandalas	Holdout p: sharp group (discovery p)
Lahiri (0)	3/3/4/4	0.40	0.44	0.039 (0.63)
Krishnamurti (−0.10°)	3/3/4/4 (no change)	0.38	0.34	0.039 (0.63)
Swamikannu Pillai (+0.03°)	3/3/4/4 (no change)	0.34	0.45	0.031 (0.63)
Fagan–Bradley (+0.88°)	3/3/4/4 (no change)	0.36	0.60	0.014 (0.41)
Raman (−1.44°)	3/5/3/3 (Sichuan, Nepal change)	0.58	0.17	0.059 (0.67)
Sepharial/traditional (−1.85°)	3/5/3/3 (Sichuan, Nepal change)	0.33	0.14	0.033 (0.64)
Chidambara Iyer (−3.2°)	3/5/2/4 (Sumatra, Sichuan, Nepal change)	0.40	0.17	0.050 (0.50)
Tropical (about +23.85°)	(see section 4)	0.41	0.29	0.705 (0.41)

Mandalas. Across the whole sweep the holdout p-values for the Parāśara Tantra and Brihat Saṃhitā mandalas range from 0.14 to 0.95, and the discovery p-values from 0.42 to 1.00. No choice of ayanāṃśa turns either into a pattern. The 15-event counts change by at most three events (3/3/4/4 under Lahiri, Krishnamurti, Swamikannu Pillai and Fagan–Bradley; 3/5/3/3 under Raman and the traditional value), which is the size of change expected from shifting the Moon by 1.4–3°, and it leaves the conclusion intact. (One of the 15 events could not be matched to the Modern M6.5+ set by time, so these counts cover 14.) **Sharp group (Ārdrā, Āśleṣā, Jyeṣṭhā, Mūla).** This is the one place where the ayanāṃśa matters for the strength of a result. On the holdout set the excess grows as the ayanāṃśa rises: 111 events against 94.5 expected at Lahiri (p=0.039), 115 at Fagan–Bradley (p=0.014) and 119 at +1.25° (p=0.005). The discovery set shows nothing at any offset (91 vs 93.6 at Lahiri; 98 vs 93.9 at +1.25°), and the pooled Modern M6.5+ set give a ratio of 1.07

(p=0.14) at Lahiri, 1.12 (p=0.039) at Fagan–Bradley and 1.15 (p=0.015) at +1.25°. **How much is chance.** Picking the best of 29 correlated offsets after seeing the holdout inflates the apparent significance. In 2,000 simulations with random times, a best-of-sweep p of 0.005 or lower occurred in 2.4% of runs and 0.011 or lower in 3.9%. So the corrected p is about 0.02–0.04, which is weak, was not pre-registered at this offset, and is absent from the discovery half.

Conclusion. (1) The headline result, no mandala or time rule separates earthquakes from chance, holds for every ayanāṃśa in the book. (2) The earlier remark that the sharp-group excess was a small-sample fluke needs softening: with 15 events it was 5 vs 2.2; with the Modern M6.5+ set it shrinks to a ratio of 1.07 at Lahiri, but it grows slightly (to 1.12–1.15) if the ayanāṃśa is set 0.9–1.25° above Lahiri. That is a lead worth one clean test, not a finding. (3) A fair confirmatory test would fix the ayanāṃśa (Lahiri or Fagan–Bradley) and the sharp-group rule in advance and use earthquakes from after October 2026, or a catalogue from before 1990. (4) Any future table should state its ayanāṃśa, since the nakṣatra, and so the mandala, of one event in nine changes between Lahiri and Raman.

12. Larger and older samples (summary; details in the Data-First Study)

The Modern M5.5+ set (9,368 earthquakes), Older set 1 (6,111, 1964–1989) and Older set 2 (670, 1900–1963) repeat the rule tests with fresh data. All numbers and methods are in the Data-First Study; this table gives the outcomes that matter for the rules in this report.

Question	Answer
Did the leads from the Modern M6.5+ set (Moon in the sharp group; Sun in Viśākhā) survive on 8,098 new earthquakes?	No: 1,176 vs 1,199 expected (0.98×, p=0.77) and 291 vs 287 (1.02×, p=0.41).
Did any of the 23 rules pass on the Modern M5.5+ set?	No. The closest, the Moon within 5° of the Sun or a planet, gave p=0.0030 against a bar of 0.0022.
Did that closest call replicate on older data?	No. Older set 1: 581 vs 594.6 (0.98×, p=0.73); Older set 2: 102 vs 99.0 (p=0.39).
Did the gap factors (formations, Rāhu, combustion, 91 two-condition pairs) pass?	No (20 primary tests, bar 0.0025; 91 pairs, bar 0.00055).
Does the data-first scan reproduce across eras?	No: r=−0.008 between the 1990–2026 and 1964–1989 scans.

13. Scope: what was tested and what was not

This table lists every kind of sky factor the texts mention, and whether it was tested in the Deadly 15 and on the larger sets (the Modern M6.5+ and M5.5+ sets; see the Data-First Study). 'Covered' means a rule was tested; 'Gap' means it has not been tested anywhere. A gap does not mean the factor is unimportant: it means no result exists for it.

Sky factor (source chapter)	Deadly 15	Larger sets	Status
Moon's star group → earthquake type (PT 18.3–18.6; BS 32.8–22)	Tested	Tested (R1; Brihat Saṃhitā version too)	Covered
Moon in a 'sharp' star group (PT ch.14)	Tested	Tested (R2, and the Data-First Study lead)	Covered
Moon's phase: tithi, karaṇa, new/full moon (PT ch.16)	Tested	Tested (R3, R4; 30 tithi bins)	Covered
Time of day (PT 18 quarters; BS 32.3–7 halves)	Tested (3 of 15 matched)	Brihat Saṃhitā version only (152 vs 159)	Partly
Sun's season: six ṛtus (PT ch.3)	Tested	Tested (R11)	Covered
Sun's star group and sign	Tested (season only)	Tested in the data-first scan (27 + 12 bins)	Covered
Eclipses (PT ch.5)	Tested (±7, 15, 30 days; 24-month rule; regions)	±7 and ±15 days only (R5, R6)	Partly: regions and 24-month rule not rerun
Comets (PT 11.19)	Tested (7 great comets)	Not rerun	Low power
Planet retrograde: Mars, Mercury, Jupiter, Venus, Saturn (PT ch.6–10)	Tested	Tested (R7)	Covered
Each planet's own star group and sign	Mars and Venus partly	Tested in the data-first scan (all 5 planets)	Covered
Moon close to the Sun or a planet (PT ch.14)	Tested (within 5°)	Tested (R10; Moon–planet angles in 30° steps)	Covered; R10 near miss did not replicate pre-1990
Sun at an angle to each planet	Not tested	Tested in the scan (30° steps)	Partly
Any two planets within 3° (PT ch.15 'war')	Tested (1° and 3°)	Tested (R9)	Covered

Sky factor (source chapter)	Deadly 15	Larger sets	Status
Jupiter and Saturn in one star group	Tested	Tested (R8)	Covered
Three or more planets in one star group (PT ch.15 formations)	Tested (0 of 15)	Tested (G1, both frames)	Covered (not confirmed)
Venus's six circles (PT ch.9)	Tested	Tested (G2, both frames)	Covered
Mars stationary; Saturn in Kṛttikā with Jupiter in Viśākhā	Tested	Mars stationary tested (G3); the Saturn–Jupiter arrangement did not occur in 1990–2026	Covered / cannot be tested
Planet-to-planet angles other than 'close' (90°, 120°, 180°)	Not tested	Tested (G4: opposition, square, trine)	Covered; specific pairs not tested
Rāhu and Ketu positions	Not tested (eclipses used instead)	Tested (G5: Rāhu's star group; Moon near the node axis)	Covered
Moon sharing a star group with planets (G6); a planet next to a particular named star	Not tested	Moon with planets tested (G6); named stars not tested (star positions not available)	Partly
Combustion (planet close to the Sun); planet rising or setting	Counted in 15 events	Combustion tested (G7); rising and setting beyond that not tested	Partly
Combined conditions (for example sharp-group Moon while Saturn is retrograde); an 'affliction score'	Not tested	Tested (G9 score; 91 two-condition pairs)	Covered
TG-SAI physics features (lunar perigee, zenith, Moon–Sun alignment)	Tested	Retrained on the Modern M6.5+ set (AUC about 0.50)	Covered
Aftershock recurrence (PT 18.7)	Tested (15 events)	Tested (481 M7+ sequences)	Covered
Omens and effects in the texts (smoke, meteors, haloes, weather, disease, regions)	Not testable	Not testable	No records exist

In short. The tests cover single factors well: the Moon's star group and phase, the Sun's season, eclipses, retrogrades, close planet pairs and Moon–planet closeness. They cover combinations poorly. The texts often describe effects when several conditions occur together (a malefic planet with the Moon in a sharp star group, a planetary formation, a planet next to a particular star). Those combinations, Rāhu and Ketu, wider planet angles, combustion and the Ch.15 formations on the large samples are the main untested areas. Each added test makes a lucky result more likely, so any extension needs rules written in advance and a stricter pass mark. The gap factors that can be tested (formations, Venus's circles, Mars standing still, wider planet angles, Rāhu, Moon with planets, combustion, an affliction score and 91 two-condition combinations) were run on the Modern M5.5+ set with rules written first (Data-First Study, section 5). None passed. A planet next to a named star and planet rising and setting remain untested.

14. Limitations: what they mean and what was done about each

In plain words. Every study has weak points. Each is listed in everyday language, with how much it matters and what was done. **Addressed** means a later test removed the concern; **Partly** means reduced; **Open** means still a limit.

Limitation	What it means in plain words	Matters	Status
Only 15 events	Fifteen quakes can reveal only a very large pattern; small ones would be missed.	High at first	**Addressed.** Repeated on the Modern M6.5+ set (1,270), the Modern M5.5+ set (9,368), Older set 1 (6,111) and Older set 2 (670).
Events chosen as the deadliest	Deaths depend on how many people lived nearby as much as on the quake, so severity tests are muddied.	Medium	**Partly.** Later samples are chosen by magnitude only, not deaths.
Rules set after seeing the 15 results	A rule tuned after peeking can fit by luck, so those results are hints, not proof.	Medium	**Addressed.** 23 rules frozen first; the last lead was re-tested with rules frozen in advance (section 12).
Baseline for slow planets	Saturn and Jupiter barely move, so comparing against all years can mislead.	Low	**Addressed.** Period-matched backgrounds and date-shift checks.
Our own sky positions	Not the commercial tool's numbers, though checked against it (largest gap 0.17°).	Low	**Addressed.** Validated on the Modern M6.5+ set (1,270 events).
One ayanāṃśa (Lahiri)	Different Indian star-measuring systems shift the star groups by up to a few degrees.	Medium	**Addressed.** Named systems and a 29-offset sweep (section 11); mandalas stay null.

Limitation	What it means in plain words	Matters	Status
Brihat Saṃhitā read from a scan	Reading Sanskrit from page images, so a misread word is possible.	Low–Medium	**Partly addressed.** Cross-checked against the English translation: all lists agree except one printing slip in the English; a Sanskrit scholar's check would still help.
Parāśara lists from an English edition	The star lists follow the translator.	Low–Medium	**Partly.** Three errors corrected against the Sanskrit (section 6.1); others could remain.
Crescent 'yoke' test uses a proxy	The Moon's tilt was approximated by its latitude, not the true horizon tilt.	Medium	**Addressed for Brihat Saṃhitā 4.12.** True-horizon version run on all eras: null. The original Parāśara 'yoke' still rests on the proxy.
Eclipse dates ±1.4 days; broad regions	Eclipse timing and ancient-region matches are fuzzy.	Medium	**Open.** Eclipse rules were not run on pre-1990 data.
Rules testable only one factor at a time	The texts often describe effects when several conditions occur together.	Medium	**Partly.** Gap tests were run (Data-First Study, section 5); none passed; some combinations never occurred.
Untestable parts	Omens, haloes, named stars, effects on crops or kings have no usable records.	Medium	**Open.** Documented in section 13, not tested.
Detects only large effects	Even 9,368 events cannot rule out an effect below about 10%.	Medium	**Partly.** Bigger samples help; a null is 'nothing found at this sensitivity'.
Data source	Only USGS was reachable; its catalogue includes ISC-GEM back to 1900, but NOAA/ISC and pre-1900 catalogues were not.	Medium	**Partly.** 1900–2026 covered; earlier centuries remain.
TG-SAI training file missing	The training file named in the repository was not in the first zip, so features were rebuilt from the repository's formulas.	Low	**Addressed.** The file was later supplied and audited (section 9.4): 104 great earthquakes, no negatives, a constant Moon-distance column and approximate planet positions; tests on it are null.

15. Follow-up items and their outcomes

Item	Outcome
Larger event sets	Done. Modern M6.5+ set (1,270), Modern M5.5+ set (9,368), Older sets 1 and 2.
Pre-registered rules and holdout	Done. 23 rules frozen; none passes on the holdout (best p=0.039 against 0.0022).
Verse 18.7: reading and completeness	Done. Completeness-corrected rerun and the Aftershock set (481 sequences, 7,323 aftershocks at M5.0): tolerances 0.25/0.5/1 day, four lag sets, both readings, M5.0 and M5.5. Observed hits at or below the decay baseline in every variant (for example 454 vs 516 expected). A Sanskrit scholar's reading of this verse would still help.
Mandala lists vs the Sanskrit	Done. Three errors in the printed PT English corrected (section 6.1); one printing slip in the BS English found (section 6.4).
Yavana sensitivity (İzmit, Turkey)	Done. İzmit's nearby eclipses fell in Śravaṇa and Āśleṣā, and Turkey 2023 had none within 30 days, so neither matches the Yavana nakṣatra (U.Phalgunī); null.
TG-SAI fixes, out-of-sample scoring and the supplied training file	Done (section 9, including 9.4): AUC about 0.50 everywhere.
Comets (PT 11.19)	Done, low power: only seven great comets in the period (section 10.3).
Historical replication (1900–1989)	Done; last open lead closed (section 12).
Crescent 'yoke'	Done. Proxy version and true-horizon version both null (section 7).
Ch.19 sky portents	Untestable: no structured records.
BS Chapter 32 effect lists	Done (section 6). Omens and regional effects cannot be tested without event records.
Ayanāṃśa sensitivity	Done (section 11). Mandalas null throughout; the sharp-group lead needs new data.
Regional belts	Done (section 8).

What remains open. A Sanskrit scholar's reading of verse 18.7 and of Brihat Saṃhitā Chapter 32, eclipse rules for older eras, and catalogues before 1900. They are unlikely to rescue a pattern that the larger tests have already ruled out at the level tested, but they are open.

16. Conclusion

The verdict. We tested the patterns in the Parāśara Tantra and the Brihat Saṃhitā, the physics measures of the TG-SAI repository, and the regional clues from Cunningham, first on the Deadly 15 and then on the Modern M6.5+ set (1,270) and Modern M5.5+ set (9,368) with rules fixed in advance. **No rule, measure or**

model predicted earthquakes better than chance.

Question	Answer	Why we trust it
Do the Deadly 15 follow the Vedic patterns?	No. 71 tests, none significant (smallest p=0.059).	A result needs p below 0.05, and below 0.0022 once 23 rules are tested together.
Do the patterns hold on thousands of earthquakes?	No. 23 rules frozen in advance; none passed on the holdout.	Tested on events never used to choose the rules.
Does the TG-SAI repository work?	No. AUC 0.50 once its flaws are fixed, and 0.50–0.51 on three older data sets it never saw.	The original high scores came from saturated features and uneven sampling.
Do the two ancient texts agree with each other?	Mostly (23 of 27 nakṣatras), and both are null.	BS read from the Sanskrit and cross-checked against the English (one printing slip, no effect on results); PT mandalas corrected.
Does the choice of ayanāṃśa change the answer?	No. Named systems and a 29-offset sweep stay null.	Section 11.

What this does and does not mean. The result is ‘no effect found at the sensitivity of these tests’. It does not prove the texts are meaningless: much of what they describe (omens, haloes, effects on crops and kings, planet risings and named stars) cannot be turned into a measurable test with the records we have (section 13), and the Deadly 15 alone is too small to detect anything but a very large effect. What it does show is that the specific, testable earthquake-timing claims do not stand up on real data, and that a computer model built on the same sky measures does no better than chance when it is built carefully.

The last lead is closed. The Moon being close to the Sun or a planet showed a faint excess in small earthquakes of the Modern M5.5+ set (p=0.0030 against a bar of 0.0022). Re-tested on Older set 1 (6,111) and Older set 2 (670), with the rule written down first, it showed no excess (0.98×, p=0.73). No lead meets the pre-registered bar in any era from 1900 to 2026 (one faint, uncorrected pattern for the crescent in Japan and Indonesia is described in section 8), and the TG-SAI model, the data-first scan and the new tests drawn from the English Brihat Saṃhitā (felt distance, time-of-day precedence, post-eclipse window) also fail on the older data.

Plain-language takeaway. If someone says the Moon's star or a planetary alignment tells you an earthquake is coming, the evidence from 1990 to 2026 does not support it. Earthquake science still rests on faults and stress, not on the sky.

17. Files

Pre-registrations: preregistration_rules.md, preregistration_bottom_up_followup.md, preregistration_m55_replication.md, preregistration_gap_tests.md, preregistration_historical_test.md, preregistration_tgsai_bottomup_historical.md, preregistration_bs_english_crosscheck.md, preregistration_bs_crescent_true_horizon.md, preregistration_regional_tests.md. **Deadly 15:** usgs_catalogue_15_events.csv, earthquakes_parashara_tantra_mandalas.csv, earthquakes_mandala_assignment.csv, tgsai_features_15_earthquakes(.csv, _extended.csv), eclipse_proximity_results.csv, eclipse_nakshatra_regions_modern.csv, eclipse_region_test.csv, ch14_16_features_15_quakes.csv, ch14_16_tests.csv, ch3_10_features_sidereal_tropical.csv, ch3_10_tests.csv. **Modern M6.5+ set:** declustered_events_M65.csv, quakes_m65_input.csv, prereg_large_sample_results.csv, tgsai_retrain_results.csv, bs_vayu_crescent_tests.csv, bs_ch32_mandala_tests.csv, ayanamsa_named_systems_tests.csv, ayanamsa_sweep_tests.csv, ch11_comet_test.csv. **Verse 18.7:** verse18_7_completeness_corrected.csv, verse18_7_M7plus_robustness.csv. **English Brihat Saṃhitā, crescent, regional:** bs_english_crosscheck_results.csv, bs_english_variants_15_events.csv, bs_crescent_true_horizon_results.csv, regional_tests_results.csv, regional_followup_date_shift.csv. **Older sets and TG-SAI scoring:** historical_replication_results.csv, historical_eraA_1964_1989_events.csv, historical_eraB_1900_1963_events.csv, historical_repo_1850_1950_events.csv, tgsai_historical_results.csv. **Supplied TG-SAI training file:** tgsai_training_file_Inputs_Mega_Quake_Topocentric_Analysis.csv, tgsai_training_file_results.csv, tgsai_training_file_refit.py, tgsai_training_file_refit_results.csv, tgsai_deadly15_rescore.csv, tgsai_deadly15_rescore.py.

18. Glossary

Term	Meaning in this context
p-value	The chance, if only luck were at work, of seeing a result at least as extreme as observed. Small means unlikely by luck alone; it is not the probability that a pattern is real.
Bonferroni threshold	The stricter p cut-off after running many tests: 0.05 divided by the number of tests (23 rules → 0.0022), so that the chance of any false positive stays at 5%.
Discovery and holdout	Two halves of the data: events before 2008 (to look for effects) and from 2008 (to confirm them). A rule counts only if it passes in the holdout and points the same way in the discovery half.
Mandala	A circle or region. In Ch.18: the four earthquake types (Anila, Anala, Indra, Varuṇa). In Ch.9: Venus's six circles. In Brihat Saṃhitā: four circles by nakṣatra.
Anila / Vāyu	Wind or gas; the earthquake type set by Moon in Svāti, Abhijit, Dhaniṣṭhā, Aśvinī, U.Phalgunī, Hasta, Citrā (corrected from the Sanskrit).
Anala / Agni	Fire; drying, heating type (Kṛttikā, Viśākhā, P.Bhādrapadā, Bharaṇī, P.Āṣāḍhā, Maghā, Puṣya, P.Phalgunī; corrected).
Indra	King of the gods; the favourable, rain-bringing type.
Varuṇa	Lord of waters; the wet, flooding type.
Nakṣatra	One of 27 equal 13°20′ divisions of the zodiac (lunar mansions), each named for a star group, e.g. Svāti (Arcturus), Rohiṇī (Aldebaran).
Sidereal / Lahiri	Zodiac fixed to the stars. Lahiri is the standard Indian ayanāṃśa used here.
Tropical	Zodiac fixed to the equinoxes/solstices; about 24° ahead of sidereal in 2000. Parāśara's solar zodiac is solstice-anchored.
Ayanāṃśa	The angular offset between the tropical and sidereal zodiacs (≈23.85° in 2000, growing about 0.014° per year).
Ṛtu / Śiśira…	Season; six of 60° each in PT Ch.3: Śiśira, Vasanta, Grīṣma, Varṣā, Śarad, Hemanta.
Tithi	Lunar day: each 12° of Moon–Sun elongation. 30 per month; PT names the 15 as Nandā, Bhadrā, Jayā, Riktā, Pūrṇā repeating.
Karaṇa	Half a tithi (6° of elongation); 11 types, with Viṣṭi regarded as harsh.
Muhūrta / Sandhyā	Thirtieth of a day (≈48 min). Sandhyās are the morning and evening twilights, when no work should begin.
Upasarga	Affliction of a star group by the Moon, a planet, a comet or a meteor (Ch.14), including occultation.
Graha-yuddha	Planetary war: close approach of two of Mars, Mercury, Jupiter, Venus, Saturn (Ch.15).
Rāhu / parvan	Rāhu is the eclipse node; a parvan is an eclipse-bearing syzygy, with seven regent deities (Ch.5).
Retrograde (vakra) / combust (asta)	Apparent backward motion of a planet / loss of visibility close to the Sun.
Syzygy	Alignment of Sun, Moon and Earth (new or full moon). The repo's feature is (1+cos 2Δ)/2 of the elongation Δ.
Perigee / zenith	Closest Moon approach / the point directly overhead; the repo uses both as 'pull' features.
Dravida, Kuru, Sauvīra, Kāśmīra, Yavana…	Ancient regions named in Ch.5 and Ch.18; mapped to modern places using Cunningham (e.g. Kāśmīra = Kashmir valley, Surāṣṭra = Kathiawar, Sauvīra = lower Indus).
TG-SAI	The repository's seven-feature earthquake model (syzygy, Saturn, Jupiter, Mars alignment, perigee, lunar and solar zenith) with a RandomForest.
Monte Carlo baseline	Repeating a test on many random dates or positions to see how often chance alone gives a result as extreme as observed.
Spearman / Kruskal–Wallis	Rank-based tests: correlation between two measures / comparison of a measure across several groups.
Declustering	Removing aftershocks so that each earthquake in a sample is an independent main shock; otherwise clusters inflate apparent patterns.
Omori's law	Aftershock rates fall roughly as 1/(time since the main shock); the natural baseline for any claimed recurrence interval.
AUC	Area under the ROC curve: 0.5 means a model cannot tell real cases from random ones; 1.0 is perfect.
ComCat	USGS's Comprehensive Earthquake Catalog, the source of the quake data used here.
Foreshock / aftershock catalogue	A list of every smaller quake before and after a main shock; needed to test verse 18.7's recurrence intervals against ordinary aftershock decay.
Pre-registration	Writing down the rules and pass marks before testing, then testing them once on data not used to choose them.
Data-set names	Deadly 15, Modern M6.5+ set, Modern M5.5+ set, Older set 1, Older set 2, Repository file and Aftershock set are defined in the table in section 1.